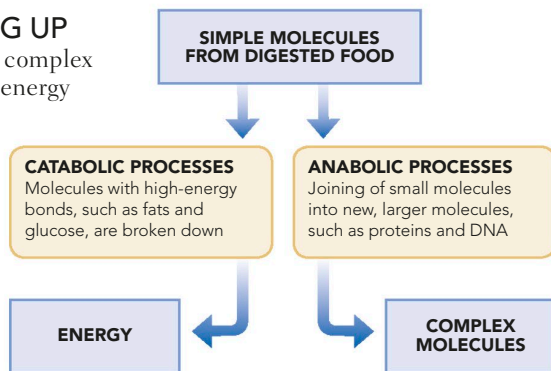


BREAKDOWN AND BUILDING UP

Catabolism is the breaking down of complex molecules into simpler ones during energy production. Anabolism is the building up of complex molecules from simpler ones – for example, amino acids make peptide chains, which combine to form proteins.

INTERPLAY

Metabolism is a complex interplay of construction and destruction, with many molecules being recycled as they pass between the two processes.



HOW THE BODY USES FOOD

The three major food components yield different breakdown products.

Carbohydrates (starches and sugars) can be reduced to glucose; proteins to single amino acids; and fats (lipids) to fatty acids and glycerol. Glucose is the body's most adaptable and readiest source of energy.

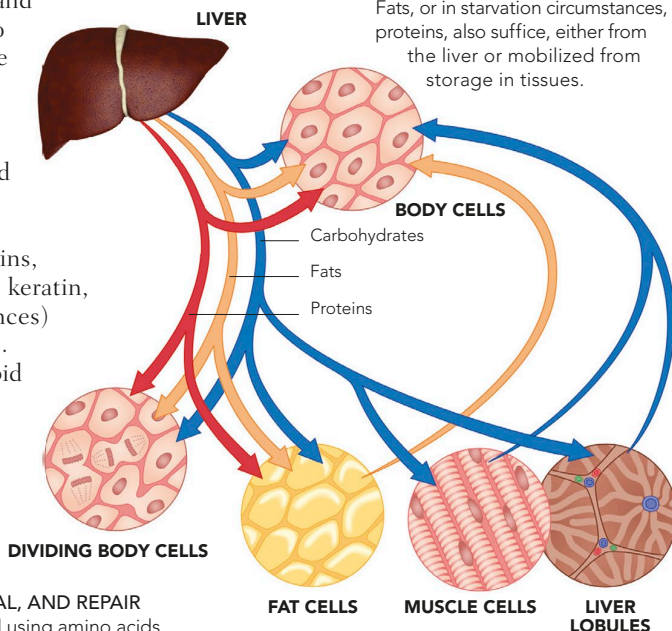
Amino acids are remade into the body's own proteins, both structural (collagen, keratin, and similar tough substances) and functional (enzymes).

Fatty acids form the bi-lipid membranes around and inside cells (see p.28).

However, the body can divert nutrients to different uses as conditions dictate.

ENERGY PRODUCTION

The simple sugar glucose is the energy source used by all cells to power their life processes. Fats, or in starvation circumstances, proteins, also suffice, either from the liver or mobilized from storage in tissues.



GROWTH, RENEWAL, AND REPAIR

Cells are maintained using amino acids to build up different protein structures, fats to form membranes, and glucose to provide the energy. Cells that are dividing for growth or repair require increased supplies of these nutrients.

ENERGY STORAGE

Surplus glucose is converted into glycogen, which is stockpiled in the liver and muscle cells. Fatty acids are a concentrated energy store, and they can be derived from dietary fats, excess amino acids, or glucose.